

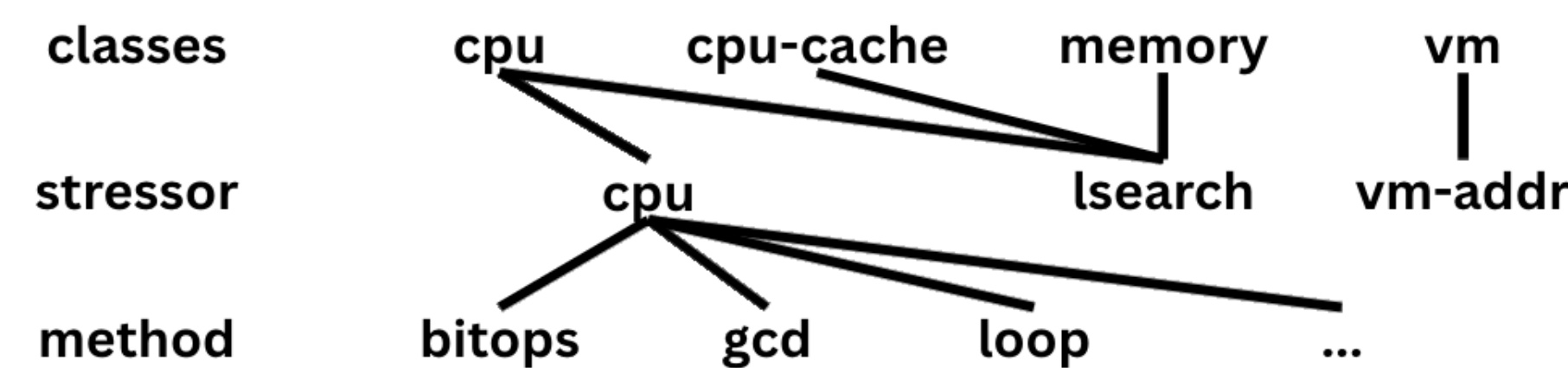
Power Output Analysis on stress-ng

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Introduction

Stress-ng is a Linux tool that stress-tests various parts of a computer system in different ways. It has hundreds of stressor options and methods to keep track of. Which combination should a system designer choose? How do they differ from one another? This lack of transparency makes it difficult to use stress-ng effectively. To address this, we coded an automated pipeline to record the total power output of various classes and methods w.r.t. CPU load and # of workers and aggregated the results.

stress-ng hierarchy example

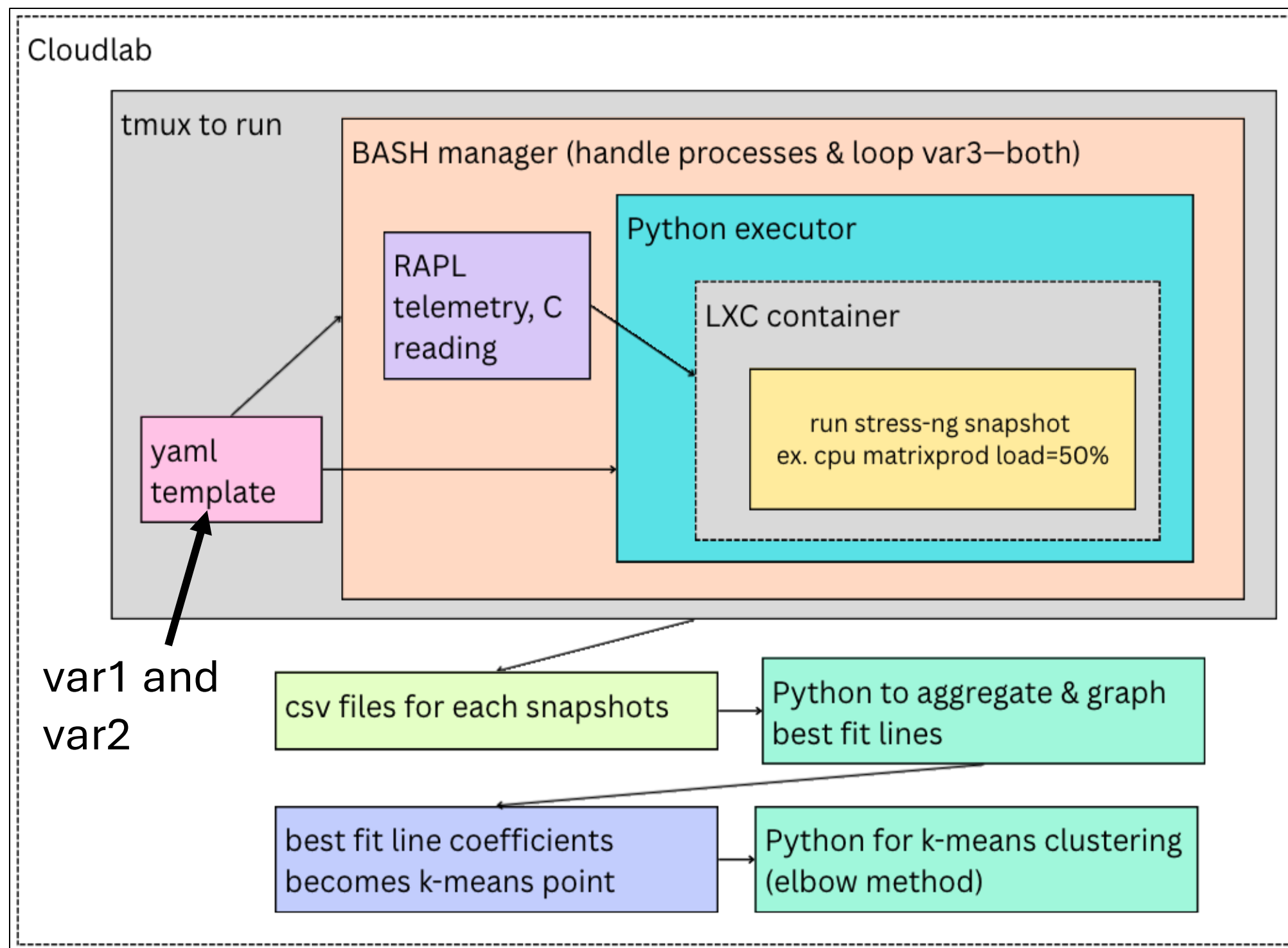


Methodology

stress-ng --cpu-load=50 --trig=8 --trig-method tanl

var1=stressor
var3=cpu load (0-100%) OR # of workers (0-40*)
var2=method

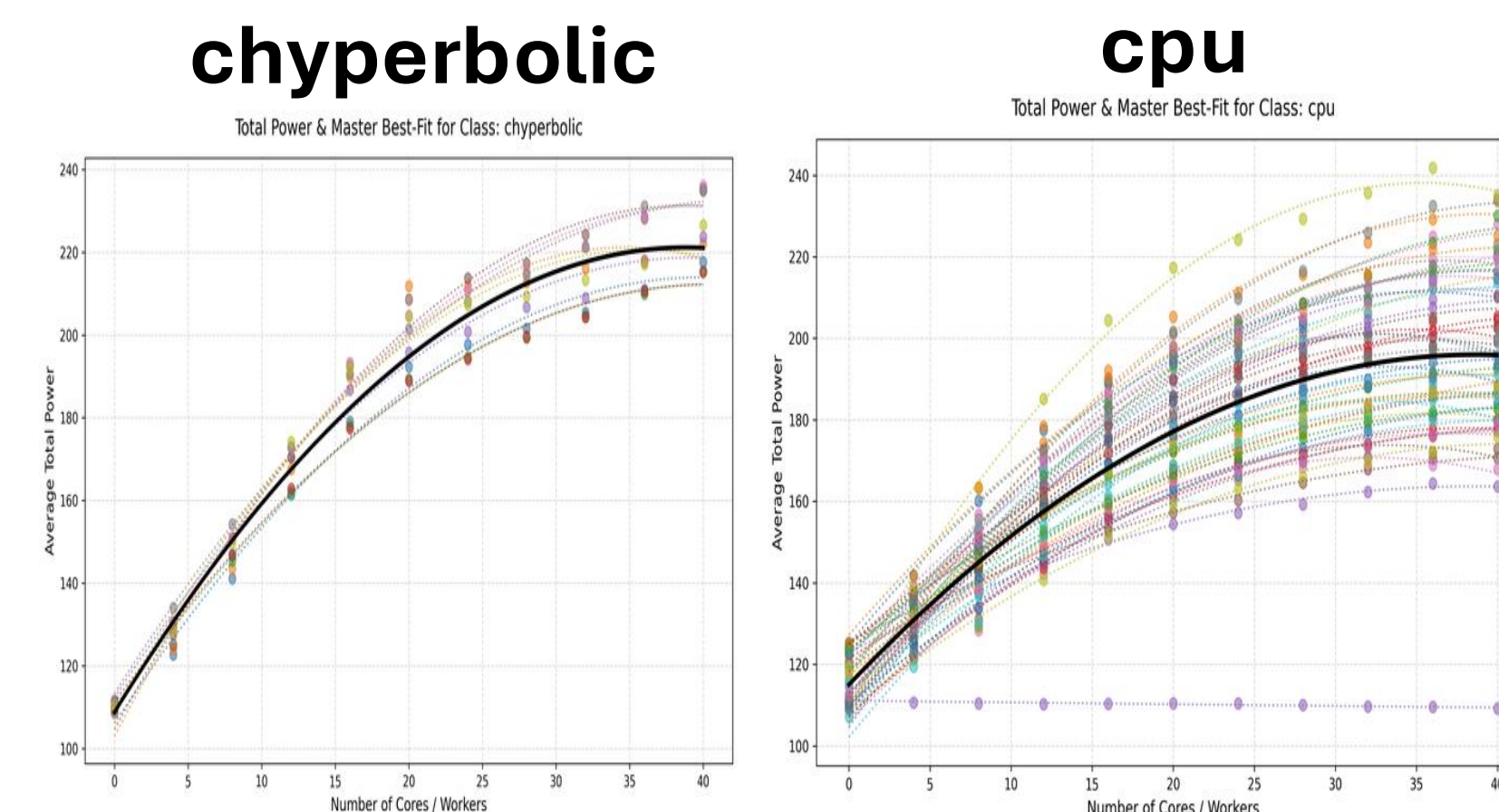
*chosen because the computer we were using had 40 max CPUs



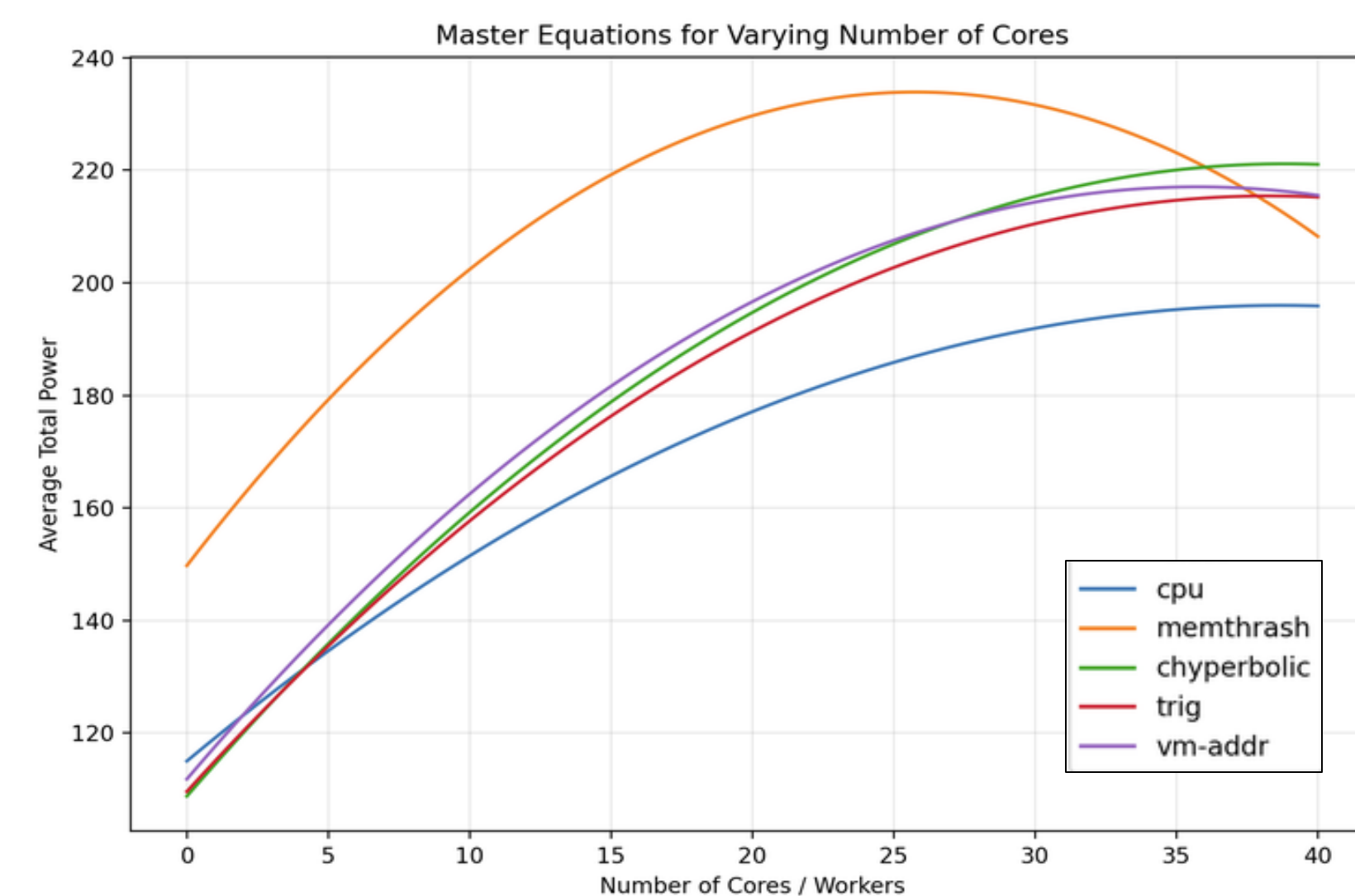
We ended up testing cpu, memthrash, chyperbolic, trig, vm-addr. Most of the available methods in those stressors were tested, but not all. cpu in particular is missing ~20 for conciseness.

Results

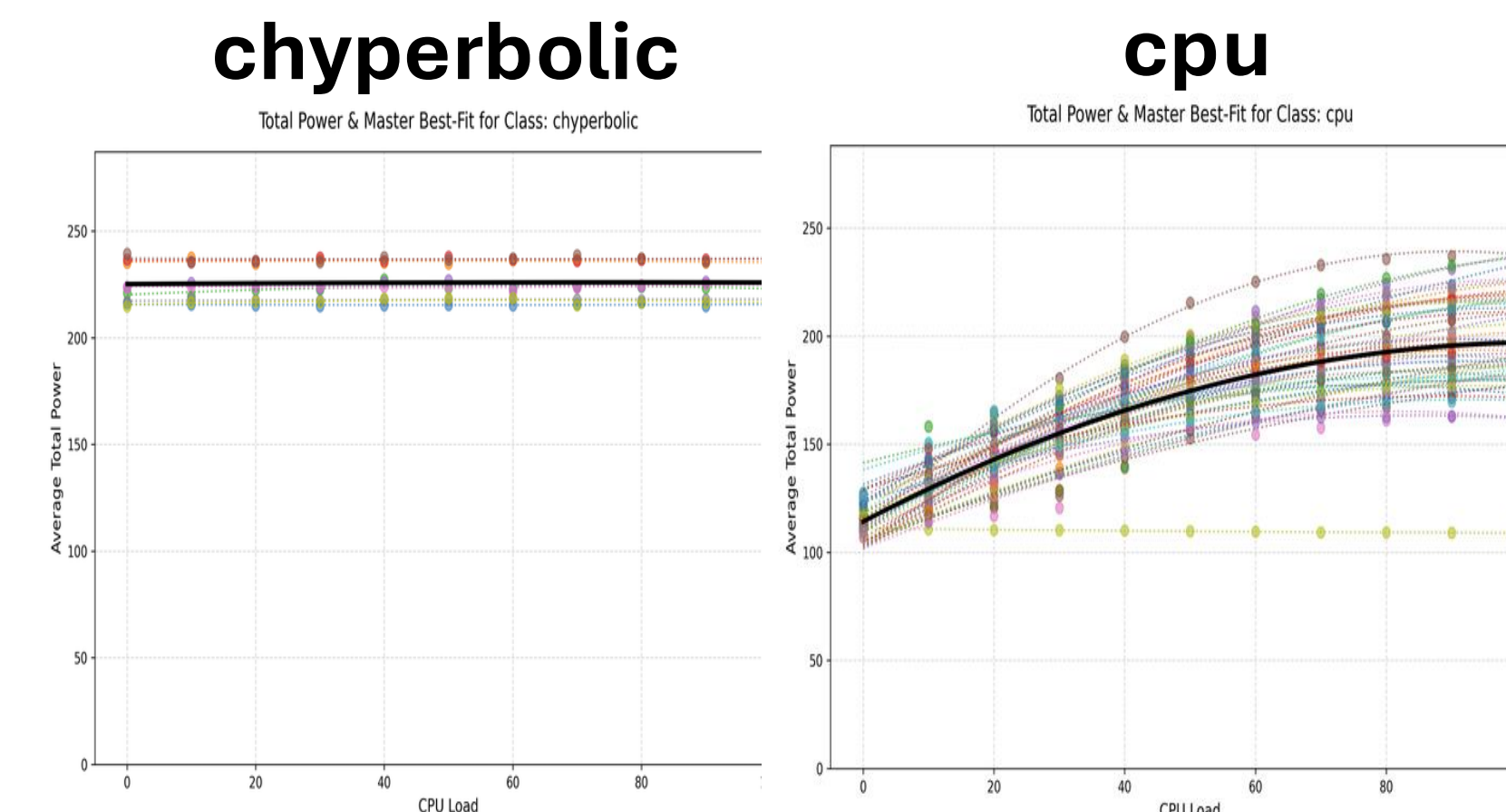
varying workers



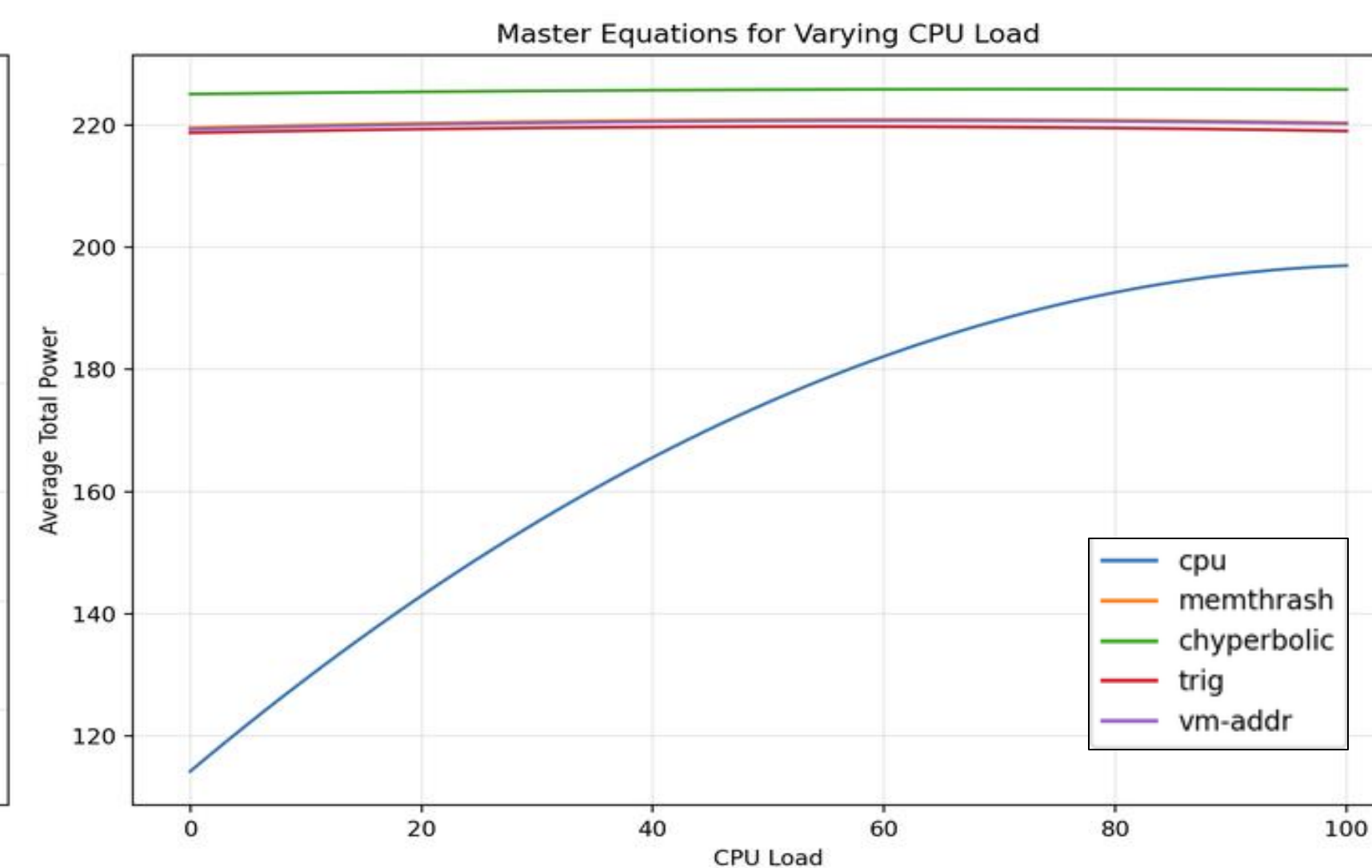
The other stressors we tested had results similar to chyperbolic and cpu. The exception was memthrash, which had a graph that spiked at 4 workers before plateauing.



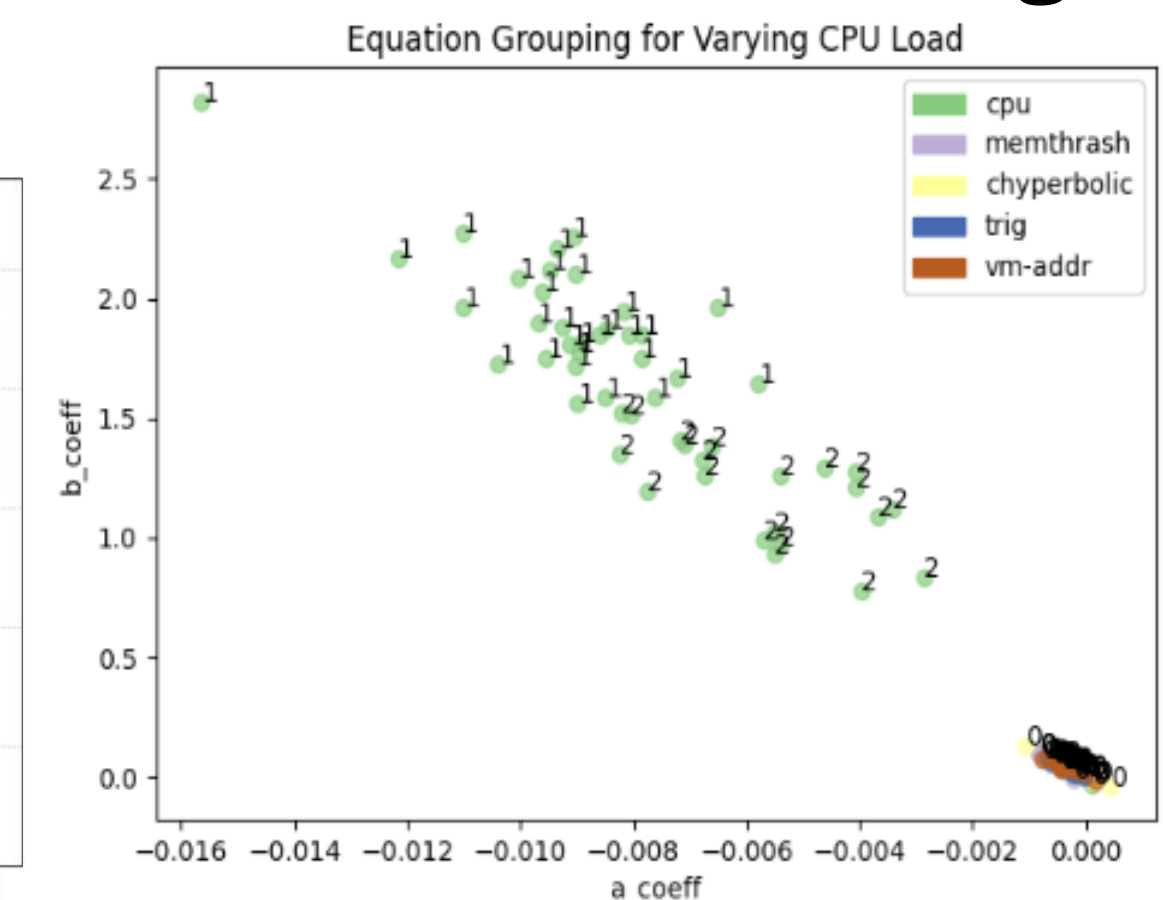
varying CPU load



All other stressors we tested matched chyperbolic—largely flat lines around 220W. Only cpu had power output scaling from increased CPU load.



k-means clustering



Both k-means for workers and CPU load had an optimal cluster number of 3. But where varying workers had no clear cluster pattern (methods of many a stressor had points across all 3 clusters), varying CPU load has a distinct divide a distinct divide between cpu and non-cpu data.

Future Work

- Further stressors (ideally of every class type)
- All stressor methods for each stressor tested
- Combinations of stressors
- Specific power readings instead of total system
- Automated yaml templates
- Accounted differences across different hardware

Conclusion

We found varying CPU load only affects certain stressors, i.e. cpu. We also found no significant patterns between classes when varying workers; we see methods of a stressor display a wide range of power output and present no strong similarity nor difference to methods of another stressor.